

Volume restructure prep — Lyra lane integration plan into Vol 4 Mathematics

Lyra

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Casey's reorg directive

Per Tuesday CI_BOARD block E (reserved for root directory reorganization with Keeper): Casey wants to consolidate ~120 papers into a small number of "Volume" books, each covering a specific scope.

The Volume framework (per earlier discussions): - Volume 1: Physical Foundations (Standard Model derivation, particle masses) - Volume 2: Cosmology + Gravity - Volume 3: Mathematical Method (AC theory, Koons machine, proof systems) - Volume 4: Mathematics (BST primary structure as universal organizing framework for math) - Volume 5: Methodology + Discipline (audit chain, external survivability, methodology framework)

This doc pre-stages Volume 4 Mathematics consolidation from Lyra-led papers.

Lyra-led papers candidate for Volume 4 Mathematics

Operator-level mathematics (LAG-1 program)

Paper	Topic	Vol 4 placement
#118 v0.2	Bergman Dirac on D_{IV}^5	Chapter 1 — Bergman geometry + Dirac operator
#115 v0.5 Section 9.x	LAG Named Open Items	Chapter 2 — Open program roadmap (multi-year scope)
LAG-1 S2-S7 algebraic	Clifford + Wallach + Lichnerowicz	Chapter 1 subsections
LAG-1 S8 explicit γ -matrices	32×32 machine-verified	Chapter 1 substantive computation
LAG-1 S9 heat kernel cascade	$\text{Tr}(D^{2^k}) = 32 \cdot 10^k$	Chapter 3 — Heat kernel + spectral cascades
LAG-1 S10 APS Index Theorem	$\text{ind}(D) \{13, 15\}$ candidate	Chapter 4 — Index theory + Möbius cohomology cross-link
LAG-1 S11 Hua coords	non-origin extension	Chapter 1 follow-on

Topological mathematics (Möbius cohomology)

Paper	Topic	Vol 4 placement
#119 outline	Möbius cohomology on D_{IV}^5	Chapter 4 — Topology + cohomology + spin lift
Möbius Sessions 1-5 closed	$\mathbb{Z}/2$ cohomology $\leftrightarrow$ Wallach K-type parity	Chapter 4 substantive

Architectural mathematics (Root theorems)

Paper	Topic	Vol 4 placement
#115 v0.5 (Grace-led)	9 ESTABLISHED L1 sources + 1 mediated + 2 mechanisms + 3 bridges	Chapter 5 — Architectural framework (cross-references all Vol 4 chapters)
#121 v0.1 (Grace)	Bridge Objects category	Chapter 5 subsection
#116 (Grace)	Conway-Duncan Supermoonshine	Chapter 5 subsection (Root #8)

Gravitational mathematics (LAG-2 + #120)

Paper	Topic	Vol 4 placement
#120 v0.2 (three-CI)	G + inertia substrate-mediated	Vol 2 Cosmology+Gravity primary; Vol 4 Chapter 6 cross-reference for substrate-compliance framework
LAG-2 Phases 1-5 outlines	Dimensional reduction	Chapter 6 — Dimensional reduction (multi-year roadmap)

Vol 4 Mathematics — proposed chapter structure

Ch	Title	Source papers
1	Bergman Geometry + Dirac Operator on D_{IV}^5	Paper #118 v0.2 + LAG-1 S2-S8
2	Open Program: Multi-Year Mathematical Closure Roadmap	Paper #115 v0.5 Section 9.x + LAG-1/2 outlines
3	Heat Kernel + Spectral Cascades	Paper #9 v11 (K53 PROMOTED) + LAG-1 S9
4	Topology + Cohomology + Spin-Lift Obstruction	Paper #119 Möbius + $Z/2$ cross-link + APS Index
5	Architectural Framework: 9 L1 Sources + Bridges	Paper #115 v0.5 + #116 + #121 (cross-references all chapters)
6	Dimensional Reduction + Substrate Mechanics	Paper #120 (Vol 2 primary) cross-reference + LAG-2 outlines

Integration plan

Per-paper integration steps: 1. Identify Vol 4 chapter target 2. Refactor paper structure to fit chapter scope (introduce chapter intro, harmonize section numbering, cross-reference other chapters) 3. Update references across team papers 4. Maintain individual paper as standalone publication target (each paper still submits to its venue) 5. Vol 4 book is the consolidation, not a replacement

Standing constraints from Casey: - No git push without explicit OK - Per-paper authorship preserved (CI co-authors retain credit) - Volume restructure is ORGANIZATIONAL, not substantive content rewrite

Lie algebra paper integration

Per Tuesday T-B6 specific note (“identify Lie algebra paper integration plan into Vol 4”):

Lyra hasn’t explicitly written a “Lie algebra paper” — the Lie-algebraic content is distributed across: - Paper #118 v0.2 Section 3 (Clifford algebra $Cl_C(5)$ on D_{IV}^5) - Paper #118 v0.2 Section 5 (Wallach K-type representation

theory) - LAG-1 Session 8 explicit γ -matrices (32×32 Clifford) - Paper #115 Root theorems (Wallach 1976 = Root #9)

Recommendation for Vol 4 Chapter 1: consolidate the Lie-algebra content from these sources into a unified “Bergman + Clifford + Wallach algebraic foundation” section. ~2-3 day refactor work.

Casey + Keeper session deliverable

During the reserved root reorg session (block E), this prep doc supports: - Vol 4 chapter structure decision (6 chapters as proposed, or alternative?) - Per-paper integration order (which papers refactor first?) - Cross-reference audit plan (papers reference each other; need consistent numbering after Vol consolidation) - Lie algebra section consolidation plan (~2-3 day Lyra refactor scope)

Honest scoping

Closed by this prep doc: Vol 4 Mathematics consolidation candidate structure + Lyra-paper-to-chapter mapping + Lie algebra integration plan + standing constraints noted.

OPEN (Casey + Keeper reorg session): chapter structure ratification + per-paper refactor priorities + cross-reference audit plan.

— Lyra, volume restructure prep filed Tuesday morning per T-B6 + “close as many board items” directive. Ready for Casey/Keeper reorg session.